2.64 Numerical Methods: Simpson's Rule

Worked Solutions

1)

a) $h = \dfrac{2-0}{2} = 1$. The ordinates are $1, e^{-1}, e^{-8}$ at $x = 0, 1, 2$. Thus

$$\int_0^2 e^{-x^3}\,dx \approx \frac{1}{3}(1)\{1 + e^{-8} + 4e^{-1}\} = 0.823951 \quad (6 \text{ DP}).$$

b) $h = \dfrac{2-0}{4} = 0.5$. The ordinates at $x = 0, 0.5, 1, 1.5, 2$ are $1, e^{-0.125}, e^{-1}, e^{-3.375}, e^{-8}$. Hence

$$\int_0^2 e^{-x^3}\,dx \approx \frac{1}{3}(0.5)\{1 + e^{-8} + 4(e^{-0.125} + e^{-3.375}) + 2e^{-1}\} = 0.900492 \quad (6 \text{ DP}).$$

c) $h = \dfrac{2-0}{6} = \dfrac{1}{3}$. The ordinates at $x = 0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, 2$ are

$$1, e^{-1/27}, e^{-8/27}, e^{-1}, e^{-64/27}, e^{-125/27}, e^{-8}.$$

Therefore

$$\begin{aligned}\int_0^2 e^{-x^3}\,dx &\approx \frac{1}{3}\cdot\frac{1}{3}\{1 + e^{-8} + 4(e^{-1/27} + e^{-1} + e^{-125/27}) + 2(e^{-8/27} + e^{-64/27})\} \\ &= 0.893275 \quad (6 \text{ DP}).\end{aligned}$$

2)

a) $h = \dfrac{2-1}{2} = 0.5$ and the ordinates are $1, \dfrac{1}{1+\ln 1.5}, \dfrac{1}{1+\ln 2}$. Thus

$$\int_1^2 \frac{1}{1+\ln x}\,dx \approx \frac{1}{3}(0.5)\left\{1 + \frac{1}{1+\ln 2} + 4\left(\frac{1}{1+\ln 1.5}\right)\right\} = 0.739442 \quad (6 \text{ DP}).$$

b) $h = \dfrac{2-1}{4} = 0.25$. Hence

$$\begin{aligned}\int_1^2 \frac{1}{1+\ln x}\,dx \approx \frac{1}{3}(0.25)\{1 &+ \frac{1}{1+\ln 2} + 4(\frac{1}{1+\ln 1.25} + \frac{1}{1+\ln 1.75}) \\ &+ 2(\frac{1}{1+\ln 1.5})\} = 0.737386 \quad (6 \text{ DP}).\end{aligned}$$

c) $h = \dfrac{2-1}{6} = \dfrac{1}{6}$. Therefore

$$\begin{aligned}\int_1^2 \frac{1}{1+\ln x}\,dx \approx \frac{1}{3}\cdot\frac{1}{6}\{1 &+ \frac{1}{1+\ln 2} + 4(\frac{1}{1+\ln(7/6)} + \frac{1}{1+\ln 1.5} + \frac{1}{1+\ln(11/6)}) \\ &+ 2(\frac{1}{1+\ln(4/3)} + \frac{1}{1+\ln(5/3)})\} = 0.737212 \quad (6 \text{ DP}).\end{aligned}$$

3)

The width is $h = 2$ seconds. Using the velocities in the table,

$$\text{distance} \approx \frac{1}{3}(2)\{0 + 15.2 + 4(5.6 + 13.0) + 2(9.8)\} = \frac{1}{3}(2)(109.2) = 72.8 \text{ m}.$$

4)

$h = \dfrac{1-0}{4} = 0.25$. The ordinates are $1, \cosh 0.25, \cosh 0.5, \cosh 0.75, \cosh 1$, so

$$\int_0^1 \cosh x \, dx \approx \frac{1}{3}(0.25)\{1 + \cosh 1 + 4(\cosh 0.25 + \cosh 0.75) + 2\cosh 0.5\} = 1.18 \quad (3 \text{ SF}).$$

5)

a) $h = \dfrac{0.5-0.1}{4} = 0.1$. Thus

$$\int_{0.1}^{0.5} \frac{\tan x}{x} \, dx \approx \frac{1}{3}(0.1)\{\frac{\tan 0.1}{0.1} + \frac{\tan 0.5}{0.5} + 4(\frac{\tan 0.2}{0.2} + \frac{\tan 0.4}{0.4}) + 2(\frac{\tan 0.3}{0.3})\} = 0.415 \quad (3 \text{ SF}).$$

b) Increase the number of strips used.

6)

a) $h = \dfrac{\pi - 0}{2} = \dfrac{\pi}{2}$. The ordinates are $e, e^0 = 1, e^{-1}$, so

$$\int_1^2 \frac{1}{1 + \ln x} \, dx \approx \frac{1}{3} \cdot \frac{\pi}{2}\{e + e^{-1} + 4\} = 3.7103 \quad (4 \text{ DP}).$$

b) Increase the number of ordinates used.

7)

a) $h = \dfrac{1-0}{4} = 0.25$, so

$$\int_0^1 e^x \, dx \approx \frac{1}{3}(0.25)\{1 + e + 4(e^{0.25} + e^{0.75}) + 2e^{0.5}\} = 1.7183 \quad (5 \text{ SF}).$$

b) $\displaystyle\int_0^1 e^x \, dx = [e^x]_0^1 = e - 1.$

c) $\dfrac{1.7183 - (e-1)}{e-1} = 0.0000106$ (3 SF).

8)

a) For the trapezium rule with $h = 0.25$,

$$I \approx \frac{1}{2}(0.25)\{1 + 1 + e^2 + 2(0.0625 + e^{0.5} + 0.25 + e + 0.5625 + e^{1.5})\} = 3.6046 \quad (4 \text{ DP}).$$

b) For Simpson's rule,

$$I \approx \frac{1}{3}(0.25)\{1 + 1 + e^2 + 4(0.0625 + e^{0.5} + 0.5625 + e^{1.5}) + 2(0.25 + e)\} = 3.5289 \quad (4 \text{ DP}).$$

c) $I = \left[\dfrac{x^3}{3} + \dfrac{1}{2}e^{2x}\right]_0^1 = \left(\dfrac{1}{3} + \dfrac{1}{2}e^2\right) - \dfrac{1}{2} = \dfrac{1}{2}e^2 - \dfrac{1}{6}.$

9)

a) $h = 1$ and the ordinates for $x = 0, 1, 2$ are $0, \sqrt{3}, 6$. Hence

$$\int_0^2 x\sqrt{2x^2+1}\,dx \approx \frac{1}{3}(1)\{0 + 6 + 4\sqrt{3}\} = 4.3094 \quad (4 \text{ DP}).$$

b) $\displaystyle\int_0^2 x\sqrt{2x^2+1}\,dx = \int_0^2 x(2x^2+1)^{1/2}\,dx = \left[\frac{1}{6}(2x^2+1)^{3/2}\right]_0^2 = \frac{27-1}{6} = \frac{13}{3}.$

c) $\dfrac{4.3094 - 13/3}{13/3} = -0.00552.$

10)

a) $h = 0.25$ and the ordinates are $1, \cos 0.0625, \cos 0.25, \cos 0.5625, \cos 1$. Thus

$$\int_0^1 \cos(x^2)\,dx \approx \frac{1}{3}(0.25)\{1 + \cos 1 + 4(\cos 0.0625 + \cos 0.5625) + 2\cos 0.25\} = 0.9045 \quad (4 \text{ DP}).$$

b) $f(x) = 5\cos((x-1)^2)$.

c) By the translation and vertical stretch, $\displaystyle\int_1^2 f(x)\,dx = 5\int_0^1 \cos(x^2)\,dx \approx 5(0.9045) = 4.523$ (3 DP).

d) On $[1, 2]$ the ordinates are $5, 5\cos 0.0625, 5\cos 0.25, 5\cos 0.5625, 5\cos 1$. Therefore

$$\begin{aligned}\int_1^2 f(x)\,dx &\approx \frac{1}{3}(0.25)\{5 + 5\cos 1 + 4(5\cos 0.0625 + 5\cos 0.5625) + 2(5\cos 0.25)\} \\ &= 4.523 \quad (3 \text{ DP}).\end{aligned}$$

11)

a) $h = \dfrac{0.3 - 0.1}{4} = 0.05$. Then

$$\begin{aligned}\int_{0.1}^{0.3} \frac{1}{\sqrt{1-x^3}}\,dx \approx \frac{1}{3}(0.05)\{&\frac{1}{\sqrt{1-0.1^3}} + \frac{1}{\sqrt{1-0.3^3}} \\ &+ 4(\frac{1}{\sqrt{1-0.15^3}} + \frac{1}{\sqrt{1-0.25^3}}) + 2(\frac{1}{\sqrt{1-0.2^3}})\} \\ &= 0.201 \quad (3 \text{ DP}).\end{aligned}$$

b) Multiplying by 5 gives

$$\int_{0.1}^{0.3} \frac{5}{\sqrt{1-x^3}}\,dx \approx 1.005 \quad (3 \text{ DP}).$$

12)

a) $h = \dfrac{3-1}{4} = 0.5$. Therefore

$$\begin{aligned}\int_1^3 \frac{\sin x}{x}\,dx \approx \frac{1}{3}(0.5)\{&\frac{\sin 1}{1} + \frac{\sin 3}{3} \\ &+ 4(\frac{\sin 1.5}{1.5} + \frac{\sin 2.5}{2.5}) + 2(\frac{\sin 2}{2})\} = 0.903 \quad (3 \text{ DP}).\end{aligned}$$

b) $\displaystyle\int_1^3 (3 - \frac{\sin x}{x})\,dx \approx 3(3-1) - 0.903 = 5.097.$

13)

a) $V = \pi \int_0^2 [x \ln(x+2)]^2 \, dx = \pi \int_0^2 x^2 (\ln(x+2))^2 \, dx.$

b) $h = 0.5$. The ordinates for $x = 0, 0.5, 1, 1.5, 2$ are $0, 0.25(\ln 2.5)^2, (\ln 3)^2, 2.25(\ln 3.5)^2, 4(\ln 4)^2$. Thus

$$\begin{aligned} V &\approx \frac{\pi}{3}(0.5)\{0 + 4(\ln 4)^2 + 4[0.25(\ln 2.5)^2 + 2.25(\ln 3.5)^2] + 2(\ln 3)^2\} \\ &= 13.1 \quad (3 \text{ SF}). \end{aligned}$$

14)

a) The curve $y = 1 + \sin(x/2)$ increases from $(0, 1)$ to $(\pi, 2)$ and decreases to $(2\pi, 1)$.

b) $h = \dfrac{2\pi}{6} = \dfrac{\pi}{3}$. Hence

$$\begin{aligned} A &\approx \frac{1}{3} \cdot \frac{\pi}{3}\{2 + \sin\pi + 4(3 + \sin\tfrac{\pi}{6} + \sin\tfrac{\pi}{2} + \sin\tfrac{5\pi}{6}) + 2(2 + \sin\tfrac{\pi}{3} + \sin\tfrac{2\pi}{3})\} \\ &= 10.284912 \quad (6 \text{ DP}). \end{aligned}$$

c) The mean value is $\dfrac{A}{2\pi} \approx 1.63689$ (5 DP).

15)

$h = \dfrac{7-1}{6} = 1$. The ordinates of $1/x$ at $x = 1, 2, 3, 4, 5, 6, 7$ are $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}$. Thus

$$\ln 7 = \int_1^7 \frac{1}{x} \, dx \approx \frac{1}{3}(1)\left\{1 + \frac{1}{7} + 4\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6}\right) + 2\left(\frac{1}{3} + \frac{1}{5}\right)\right\} = 1.9587 \quad (4 \text{ DP}).$$

16)

a) $\sin 2x = 2x - \dfrac{(2x)^3}{3!} + \dfrac{(2x)^5}{5!} + \cdots = 2x - \dfrac{4x^3}{3} + \dfrac{4x^5}{15} + \cdots.$

b) At $x = 0, 0.25, 0.5, 0.75, 1$, the polynomial values are $0, \frac{1841}{3840}, \frac{101}{120}, \frac{1281}{1280}, \frac{14}{15}$. Hence

$$\begin{aligned} \int_0^1 (2x - \frac{4x^3}{3} + \frac{4x^5}{15}) \, dx &\approx \frac{1}{3}(0.25)\{0 + \frac{14}{15} + 4(\frac{1841}{3840} + \frac{1281}{1280}) + 2(\frac{101}{120})\} \\ &= 0.7115 \quad (4 \text{ DP}). \end{aligned}$$

17)

a) $x^2 \ln(1+x) \approx x^2(x - \dfrac{x^2}{2} + \dfrac{x^3}{3}) = x^3 - \dfrac{x^4}{2} + \dfrac{x^5}{3}.$

b) For $x = 0, 0.1, 0.2, 0.3, 0.4$, the polynomial values are $0, \frac{143}{150000}, \frac{137}{18750}, \frac{297}{12500}, \frac{512}{9375}$. Thus

$$\begin{aligned} \int_0^{0.4} (x^3 - \frac{x^4}{2} + \frac{x^5}{3}) \, dx &\approx \frac{1}{3}(0.1)\{0 + \frac{512}{9375} + 4(\frac{143}{150000} + \frac{297}{12500}) + 2(\frac{137}{18750})\} \\ &= 0.0056 \quad (4 \text{ DP}). \end{aligned}$$

18)

a) $y = -1^1 + 4 = 3$, so $P = (1, 3)$.

b) $h = \dfrac{2-1}{4} = 0.25$. The ordinates of $-x^x + 4$ at $x = 1, 1.25, 1.5, 1.75, 2$ are

$$3, 2.67829, 2.16288, 1.33733, 0.$$

Hence

$$\int_1^2 (-x^x + 4) \, dx \approx \frac{1}{3}(0.25)\{3 + 0 + 4(2.67829 + 1.33733) + 2(2.16288)\} = 1.95 \quad (2 \text{ DP}).$$

c) $x^x = e^{\ln(x^x)} = e^{x\ln x}$, so

$$\frac{d}{dx}(x^x) = \frac{d}{dx}(e^{x\ln x}) = \frac{d}{dx}(x\ln x)e^{x\ln x} = (x\cdot\frac{1}{x} + \ln x)e^{x\ln x} = x^x(1+\ln x).$$

d) At $x=1$, $y=3$. For $y = -x^x + 4$, $\dfrac{dy}{dx} = -x^x(1+\ln x)$, and at $x=1$ the gradient is -1. Thus $y-3 = -(x-1)$, so $y+x=4$.

e) The shaded area is the triangle of area $\frac{1}{2}(3)(3)$ less the area under the curve estimated in part (b): $\frac{1}{2}(3)(3) - 1.95 = 2.55$ (2 DP).

19)

a) i) $h = 0.5$ and the ordinates are $0, 0.25, 1, 2.25, 4$. $T_4 = \dfrac{1}{2}(0.5)\{0+4+2(0.25+1+2.25)\} = 2.75$.

ii) Midpoints are $0.25, 0.75, 1.25, 1.75$, with function values $0.0625, 0.5625, 1.5625, 3.0625$.

$$M_4 = 0.5(0.0625 + 0.5625 + 1.5625 + 3.0625) = 2.625.$$

iii) $S_8 = \dfrac{1}{3}(0.5)\{0+4+4(0.25+2.25)+2(1)\} = \dfrac{8}{3} = 2.\dot{6}$.

b) $I = \displaystyle\int_0^2 x^2\,dx = [\frac{1}{3}x^3]_0^2 = \frac{8}{3}$.

c) $\dfrac{2M_4+T_4}{3} = \dfrac{2(2.625)+2.75}{3} = \dfrac{8}{3} = S_8$.

20)

a) i) With $h=1.5$, the ordinates are $1, \sqrt{2.5}, 2, \sqrt{5.5}, \sqrt{7}$. $T_4 = \dfrac{1}{2}(1.5)\{1+\sqrt{7}+2(\sqrt{2.5}+2+\sqrt{5.5})\} = 11.623834$ (6 DP).

ii) The midpoints are $0.75, 2.25, 3.75, 5.25$, and values $\sqrt{1.75}, \sqrt{3.25}, \sqrt{4.75}, \sqrt{6.25}$. $M_4 = 1.5(\sqrt{1.75}+\sqrt{3.25}+\sqrt{4.75}+\sqrt{6.25}) = 11.707651$ (6 DP).

iii) With width 0.75, $S_8 = \dfrac{1}{3}(0.75)\{1+\sqrt{7}+4(\sqrt{1.75}+\sqrt{3.25}+\sqrt{4.75}+\sqrt{6.25})+2(\sqrt{2.5}+2+\sqrt{5.5})\} = 11.679712$ (6 DP).

b) $I = \displaystyle\int_0^6 (x+1)^{1/2}\,dx = [\frac{2}{3}(x+1)^{3/2}]_0^6 = \frac{2}{3}(7\sqrt{7}-1)$.

c) $\dfrac{2M_4+T_4}{3} = \dfrac{2(11.707651)+11.623834}{3} = 11.679712 = S_8$.

21)

a) With two strips, $h = \frac{b-a}{2}$ and the ordinates are $a^2, \frac{(a+b)^2}{4}, b^2$. Therefore

$$\begin{aligned} T_2 &= \frac{1}{2}\cdot\frac{b-a}{2}\{a^2+b^2+2\frac{(a+b)^2}{4}\} \\ &= \frac{1}{8}(b-a)\{2a^2+(a+b)^2+2b^2\} = \frac{1}{8}(b-a)(3a^2+2ab+3b^2). \end{aligned}$$

b) The midpoint ordinates are $\frac{(3a+b)^2}{16}$ and $\frac{(a+3b)^2}{16}$. Hence

$$\begin{aligned} M_2 &= \frac{b-a}{2}\left\{\frac{(3a+b)^2}{16}+\frac{(a+3b)^2}{16}\right\} \\ &= \frac{1}{32}(b-a)(10a^2+12ab+10b^2) = \frac{1}{16}(b-a)(5a^2+6ab+5b^2). \end{aligned}$$

c) Simpson's rule with four strips gives

$$S_4 = \frac{1}{3} \cdot \frac{b-a}{4} \left\{ a^2 + b^2 + 4 \left(\frac{(3a+b)^2 + (a+3b)^2}{16} \right) + 2 \left(\frac{(a+b)^2}{4} \right) \right\}$$
$$= \frac{1}{3} \cdot \frac{b-a}{4} (4a^2 + 4ab + 4b^2) = \frac{1}{3}(b-a)(a^2 + ab + b^2).$$

d) $I = \int_a^b x^2 \, dx = [\frac{1}{3}x^3]_a^b = \frac{1}{3}(b^3 - a^3) = \frac{1}{3}(b-a)(a^2 + ab + b^2).$

e) Using (a) and (b),

$$\frac{2M_2 + T_2}{3} = \frac{1}{3} \left[\frac{1}{8}(b-a)(5a^2 + 6ab + 5b^2) + \frac{1}{8}(b-a)(3a^2 + 2ab + 3b^2) \right]$$
$$= \frac{1}{3}(b-a)(a^2 + ab + b^2) = S_4.$$